RIPHAH INTERNATIONAL UNIVERSITY

Faculty of Computing

FINAL YEAR PROJECT PROPOSAL & PLAN

Workforce Monitoring for Corporate Cleaning Services

Project Team

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[Project Supervisor]

(SYED SHARJEEL GILANI)

Workforce Monitoring for Corporate Cleaning Services

Change Record

Author(s)	Version	Date	Notes	Supervisor's Signature
Shumail	1.0	22/2/2025	Original Draft	
			Changes Based on Feedback From Supervisor	
			Changes Based on Feedback From Faculty	
			Added Project Plan	
			Changes Based on Feedback From Supervisor	

Project Proposal

Project Title Workforce Monitoring for Corporate Cleaning Services

Opportunity & Stakeholders:

Traditional cleaning management is reactive, causing inefficiencies and human error, while occupational safety suffers from limited real-time hazard detection. Leveraging AI, this project digitizes workforce oversight to improve productivity and transparency.

Existing System/ Description of the Current Situation:

- **Reactive Management:** Monitoring is manual and delayed, preventing proactive interventions for efficiency and safety.
- **Manual Interference:** Reliance on third-party intermediaries causes delays in communication and increases data security risks.
- **Lack of Integration:** No unified system combines real-time tracking, task verification, and manager feedback.
- **Limited Transparency:** Current processes do not provide clear oversight of worker performance or task completion.

Problem Statement:

Current workforce management relies on third-party platforms, lacks real time monitoring, direct communication, and transparent performance tracking, and depends on manual attendance and reporting.

Proposed Solution:

- Replace manual tracking with a digital, data-driven workforce management system.
- Continuously monitor task progress, worker location, and workplace conditions.

- Use predictive analytics to detect potential performance drops.

- Trigger adaptive feedback and alerts for missed tasks or late arrivals, enabling immediate corrective actions.

Scope of the Project:

- **Staff Side:** Enable real-time location tracking, digital task management, photo-based task verification, and emergency incident reporting.

- **Manager Side:** Provide dashboards for performance analytics, automated attendance tracking, and direct task assignment and communication tools.

- **AI Module:** Implement AI driven verification of work quality using an image-classification model.

List of Faculty Proposed Changes

Workforce Monitoring for Corporate Cleaning Services

Supervisor's Signature: _____

Proposed Change	**Proposed By**	**Supervisor's Decision**
	Name of Faculty Member(s) who proposed this change	Approved/Disapproved and/or Comments

Project Plan

Work Breakdown Structure: A work breakdown structure (WBS) is deliverable based decomposition of project scope. The WBS includes 100% of the work defined by the project scope and captures all deliverables – internal, external, interim – in terms of the work to be completed, including project management.

1. **Project Management**
   - 1.1. Work Breakdown Structure (WBS)
   - 1.2. Roles & Responsibility Matrix
   - 1.3. Change Control System
2. **Reports / Documentation**
   - 2.1. Final Documentation Introduction
   - 2.2. Literature / Markey Survey
   - 2.3. Requirements Analysis
   - 2.4. System Design
   - 2.5. Implementation
   - 2.6. Testing & Performance Evaluation
   - 2.7. Conclusion & Outlook
   - 2.8. End User Documentation
   - 2.9. Application Administration Documentation
   - 2.10. System Administrator Documentation
3. **System**
   - 3.1 Development Environment
     - 3.1.1 VS Code / IDE Setup
     - 3.1.2 Git & GitHub Version Control
     - 3.1.3 Node.js / Express Server
     - 3.1.4 MongoDB Atlas Database
   - 3.2 Presentation Layer
     - 3.2.1 Staff Login & Signup Screens
     - 3.2.2 Staff Dashboard (Task List, Schedule)
     - 3.2.3 Task Detail & Photo Upload Screen
     - 3.2.4 Real-time Location Tracking UI
     - 3.2.5 Attendance Check-in / Check-out Screen

3.2.6 Emergency Incident Reporting Screen
3.2.7 Notifications & Alerts Screen
3.2.8 Manager Dashboard (KPIs, Charts, Overview)
3.2.9 Staff Management Page
3.2.10 Task Management Page
3.2.11 Live Tracking Map View
3.2.12 Attendance & Shift Overview Page
3.2.13 Performance Reports & Analytics Page
3.2.14 Direct Communication / Messaging Module
3.3 Business Logic Layer
3.3.1 Authentication & RBAC Middleware
3.3.2 Task Template & Instance Generation Logic
3.3.3 Task Assignment & Scheduling Logic
3.3.4 Automated Attendance Tracking Logic
3.3.5 Real-time Location Tracking Integration
3.3.6 Photo-based Task Verification Logic
3.3.7 Notification & Alert Trigger System
3.3.8 Performance Analytics & KPI Calculations
3.3.9 Adaptive Feedback & Smart Alerts
3.3.10 AI Cleaning Verification Model Integration
3.4 Data Management Layer
3.4.1 User & Profile Data Models
3.4.2 Site / Location Data Models
3.4.3 Task Template & Instance Data Models
3.4.4 Attendance & Shift Data Models
3.4.5 Incident Report Data Models
3.4.6 Performance & Analytics Data Models
3.5 Physical Layer
3.5.1 MongoDB Atlas Cloud Database Deployment
3.5.2 Cloud Server Deployment (Backend API)
3.5.3 Frontend App Deployment / Publishing
3.5.4 AI Model Hosting / Inference Server

Roles & Responsibility Matrix:

The purpose of roles & responsibility matrix is to identify who will do what.

WBS #	WBS Deliverable	Activity #	Activity to Complete the Deliverable	Duration (# of Days)	Responsible Team Member(s) & Role(s)
1	**Project Planning & Requirement Analysis**	1.1	Gather system requirements, stakeholder interviews & finalize project scope	7	All Members
		1.2	Define functional & non-functional requirements (SRS Document)	5	Umer (Lead), Shumail
		1.3	Feasibility study & technology stack selection	4	Shumail (Lead), Ali
		1.4	Create Work Breakdown Structure & project timeline	3	All Members
2	**System Design**	2.1	Design overall system architecture (Frontend + Backend + AI pipeline)	5	Shumail (Lead), All Members
		2.2	Design database schema and data models	5	Shumail
		2.3	Create UI/UX wireframes for Staff App modules	5	Ali (Lead), Umer
		2.4	Create UI/UX wireframes for Manager Dashboard modules	5	Ali (Lead), Umer
		2.5	Design API architecture & endpoint contracts	4	Shumail
		2.6	Design AI model pipeline for cleaning verification	4	Shumail (Lead), Ali
3	**Frontend Development Staff App**	3.1	Setup frontend project structure & design system	4	Ali (Lead), Umer
		3.2	Develop Login & Signup screens	4	Ali, Umer
		3.3	Develop Staff Dashboard (task list, schedule view)	6	Ali (Lead), Umer
		3.4	Develop Task Detail & Photo Upload screen	5	Ali
		3.5	Develop Real-time Location Tracking UI	5	Ali (Lead), Umer
		3.6	Develop Attendance Check-in / Check-out screen	4	Umer

		3.7	Develop Emergency Incident Reporting screen	3	Umer
		3.8	Develop Notifications & Alerts screen	4	Ali
4	**Frontend Development Manager Dashboard**	4.1	Develop Manager Login & Role-based Routing	3	Ali, Umer
		4.2	Develop Manager Dashboard (overview, KPIs, charts)	7	Ali (Lead), Umer
		4.3	Develop Staff Management page (assign, view, edit staff)	5	Umer (Lead), Ali
		4.4	Develop Task Management page (create, assign, track tasks)	6	Ali (Lead), Umer
		4.5	Develop Live Tracking Map view	6	Ali
		4.6	Develop Attendance & Shift Overview page	5	Umer
		4.7	Develop Performance Reports & Analytics page (with charts)	7	Ali (Lead), Umer
		4.8	Develop Direct Communication / Messaging module	5	Umer
5	**Backend Development Core**	5.1	Setup backend environment, project structure & dev tools	3	Shumail
		5.2	Implement authentication system (JWT, Login/Signup for Staff & Manager)	6	Shumail (Lead), Umer
		5.3	Implement role-based access control (RBAC) middleware	4	Shumail
		5.4	Develop User & Profile management APIs	5	Shumail, Umer
		5.5	Develop Site/Location management APIs	5	Shumail
6	**Backend Development Task & Scheduling**	6.1	Develop Task Template CRUD APIs (recurring tasks)	7	Shumail
		6.2	Develop Task Instance generation logic (daily/weekly/monthly)	6	Shumail
		6.3	Develop Task Assignment & Reassignment APIs	5	Shumail, Umer
		6.4	Develop Scheduling & Shift management APIs	6	Umer (Lead), Shumail
7	**Backend Development Monitoring & Tracking**	7.1	Implement automated attendance tracking APIs	7	Shumail (Lead), Ali
		7.2	Develop real-time location tracking API	6	Ali (Lead), Shumail

			integration		
		7.3	Develop photo-based task verification upload & storage APIs	6	Shumail
		7.4	Develop emergency incident reporting APIs	4	Umer
		7.5	Implement notification & alert system	5	Umer (Lead), Ali
8	**Backend Development Analytics & Reporting**	8.1	Develop performance analytics & KPI calculation APIs	7	Shumail
		8.2	Develop reporting endpoints (daily/weekly/monthly reports)	5	Umer (Lead), Shumail
		8.3	Implement adaptive feedback & smart alert triggers	5	Shumail, Ali
9	**Data Integrity & Upload Validation**	9.1	Implement upload validation and integrity checks for submitted photos	5	Shumail
		9.2	Develop mechanisms to ensure authenticity of task evidence	5	Shumail (Lead), Ali
		9.3	Implement client-side upload restrictions and safeguards	3	Ali
10	**AI Module Development**	10.1	Research & select suitable image classification approach for cleaning verification	7	Shumail (Lead), Ali
		10.2	Collect & label training dataset (before/after cleaning images)	10	All Members
		10.3	Train & fine-tune the AI model	10	Shumail (Lead), Ali
		10.4	Develop AI inference API (model serving endpoint)	5	Shumail
		10.5	Integrate AI verification with task completion workflow	5	Shumail, Ali
		10.6	Test & validate AI model accuracy and edge cases	7	All Members
11	**Integration & System Testing**	11.1	Integrate Staff App frontend with backend APIs	7	Ali, Umer
		11.2	Integrate Manager Dashboard frontend with backend APIs	7	Ali, Umer
		11.3	End-to-end integration testing	7	All Members
		11.4	Performance & load testing	5	Shumail (Lead), Umer
		11.5	Security testing & vulnerability	4	Shumail

			assessment		
		11.6	User acceptance testing (UAT) with sample users	5	All Members
		11.7	Bug fixing & optimization iteration	10	All Members
12	**Documentation**	12.1	Write Literature Review / Market Survey	7	Umer (Lead), Ali
		12.2	Write Requirements Analysis chapter	5	Umer
		12.3	Write System Design chapter (architecture, ERD, DFD, UML)	7	Shumail (Lead), Umer
		12.4	Write Implementation chapter	6	All Members
		12.5	Write Testing & Performance Evaluation chapter	5	Umer (Lead), Shumail
		12.6	Write Conclusion & Future Work chapter	3	Umer
		12.7	Prepare End-User Documentation / User Manual	5	Ali (Lead), Umer
		12.8	Prepare System & Application Admin Documentation	4	Shumail
		12.9	Final report compilation, formatting & review	5	All Members
13	**Deployment & Presentation**	13.1	Deploy backend to cloud server (production)	4	Shumail
		13.2	Deploy frontend / publish app	3	Ali (Lead), Shumail
		13.3	Prepare final demo & presentation slides	5	All Members
		13.4	Rehearsal & dry run for FYP defense	3	All Members
		13.5	FYP Final Defense / Viva	1	All Members

Approval

Project Supervisor

Comments

Name:

Date: **Signature:**

Project Coordinator

Comments

Name:

Date: **Signature:**